

01. Resources and Development

01. What is a resource?

Everything available in our environment which can be used to satisfy our needs, provided, it is technologically accessible, economically feasible and culturally acceptable can be termed as 'Resource'.

02. Name the factors involved in producing resources.

- [i] Nature
- [ii] Technology
- [iii] Institution

Human beings interact with nature through technology and create institutions to form resources.

03. Do you think that resources are free gifts of nature? Give examples.

No, resources are not free gifts of nature because human beings put their money and labor/effort in order to make things resources. They are not resources by their own. Example – cotton. We obtain cotton, process it using money and labor and make shirts, etc. They are now resource.

04. Classify resources into different categories.

These resources can be classified in the following ways–

- (a) On the basis of origin – biotic and abiotic
- (b) On the basis of exhaustibility – renewable and non-renewable
- (c) On the basis of ownership – individual, community, national and international
- (d) On the basis of status of development – potential, developed stock and reserves.

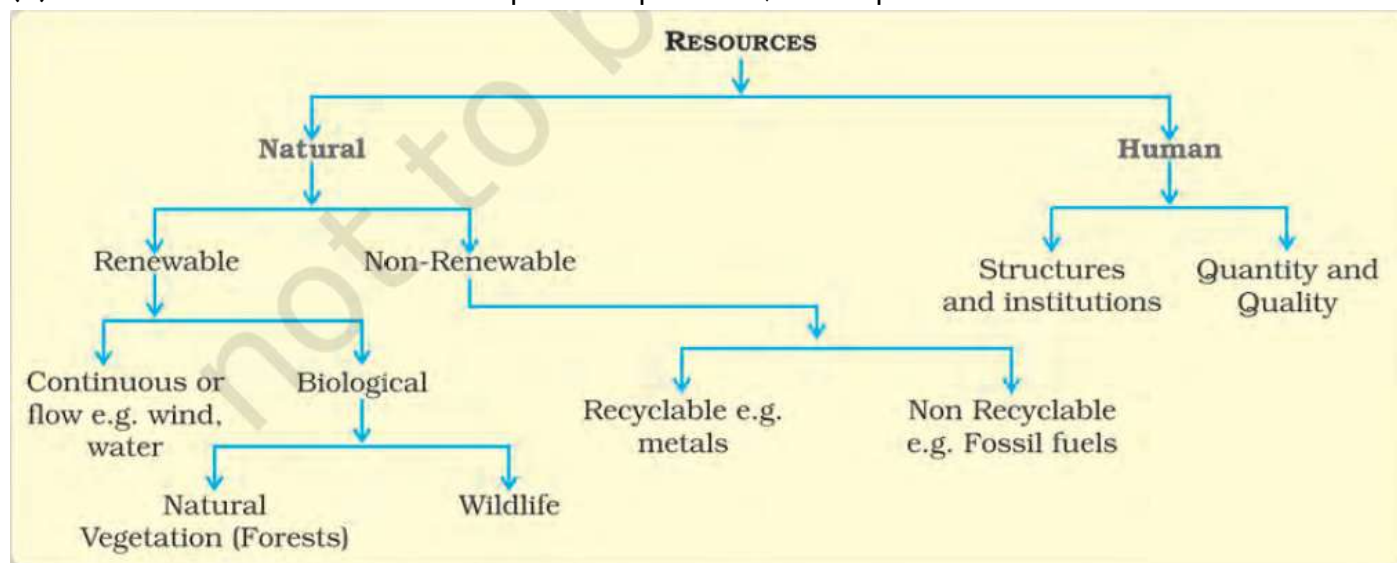


Fig 1: Classification of resources

05. What are the bad effects of developments of resources?

The bad effects of developments of resources are:

- Depletion of resources for satisfying the greed of a few individuals.
- Accumulation of resources in few hands, which, in turn, divided the society into two segments i.e. haves and have nots or rich and poor.
- Indiscriminate exploitation of resources has led to global ecological crises such as, global warming, ozone layer depletion, environmental pollution and land degradation.

06. Why should we conserve resources?

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- Stop Irrational Consumption and over-utilization: Indiscriminate use of resources has led to socio-economic problems and environmental crisis.
- Promote Sustainability for Future Generations: Resources are vital for any developmental activity, but since many are non-renewable, we must use them wisely and save for future generations.
- Stop Inequitable distribution: Resources are accumulated in a few hands, dividing society into 'haves' and 'have nots'; conservation helps ensure equitable distribution and global peace.

07. How should you conserve resources?

We can conserve resources by:

- Using resources wisely – Resource planning should be adopted to ensure the judicious and sustainable use of resources in a diverse country like India.
- Following the 3R's – Reducing, reusing and recycling helps minimize waste and lessens the burden in our finite natural reserves.
- Using non-conventional sources of energy – Shifting to renewable sources of energy like solar or wind energy.
- Minimizing mining activities – Controlled mining is necessary to prevent the rapid depletion of minerals and to protect the land from degradation.
- Conserving the environment – Preventing exploitation maintains the ecological balance, ensuring that life-supporting systems like air, water and soil remain functional.

08. What is resource planning?

An equitable resource planning is distribution of resources and a widely accepted strategy for judicious use of resources.

09. Why should a country like India adopt resource planning?

It has importance in a country like India, which has enormous diversity in the availability of resources. There are regions which are rich in certain types of resources but are deficient in some other resources. There are some regions which can be considered self-sufficient in terms of the availability of resources and there are some regions which have acute shortage of some vital resources. Arunachal Pradesh has abundance of water resources but lacks in infrastructural development. The state of Rajasthan is very well endowed with solar and wind energy but lacks in water resources. This calls for balanced resource planning at the national, state, regional and local levels.

10. Mention the different steps involved in resource planning?

Resource planning is a complex process which involves:

- [i] Identification and inventory of resources across the regions of the country. This involves surveying, mapping and qualitative and quantitative estimation and measurement of the resources.
- [ii] Evolving a planning structure endowed with appropriate technology, skill and institutional set up for implementing resource development plans.
- [iii] Matching the resource development plans with overall national development plans.

11. Can you name some resource rich but economically backward regions and some resource poor but economically developed regions? Give reasons for such a situation.

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- Resource rich but backward: Jharkhand and Chhattisgarh. Rich in minerals but lack in infrastructure.
- Resource poor but developed: Punjab and Haryana. Lacking in minerals but agriculturally and technologically advanced.
Reason: Development requires not just resources, but also technology, quality and human resources, and institutional changes.

12. How much of the Indian land mass is covered by plains, mountains and plateaus?

Mention their importance.

India possesses a very diverse physical geography. The land mass is divided into the following approximate proportions:

- Plains (43%) – About 43% of the Indian land mass is covered with plains.
- Mountains (30%) – About 30% of the Indian land mass is covered with mountains.
- Plateaus (27%) – About 27% of the Indian land mass is covered with plateaus.

Importance –

- [i] Plains – The largest portion of the country is flat, fertile land, making them “granary of India”. Being flat, these areas are ideal for setting up industries, connecting transport network and supporting high population densities.
- [ii] Mountains – High mountain ranges like the Himalayas are the source of major perennial rivers and it also provides sites for tourism and ecological balance.
- [iii] Plateau – The Peninsular Plateau like the Chota Nagpur region is a power-house of minerals including iron, coal and manganese. It also has vast forest resources which are rich in raw materials like timber and paper.

13. Mention the different purpose for which land is utilized.

Land resources are used for the following purposes:

- [i] Forests
- [ii] Land not available for cultivation
 - (a) Barren and waste land
 - (b) Land put to non-agricultural uses, e.g. buildings, roads, factories, etc.
- [iii] Other uncultivated land (excluding fallow land)
 - (a) Permanent pastures and grazing land,
 - (b) Land under miscellaneous tree crops groves (not included in net sown area),
 - (c) Culturable waste land (left uncultivated for more than 5 agricultural years).
- [iv] Fallow lands
 - (a) Current fallow – (left without cultivation for one or less than one agricultural year),
 - (b) Other than current fallow – (left uncultivated for the past 1 to 5 agricultural years).
- [v] Net sown area – the physical extent of land on which crops are sown harvested is known as net sown area.

14. Mention the reasons of land degradation.

The reasons of land degradation are as follows:

- Human activities such as deforestation, over grazing, mining and quarrying too have contributed significantly in land degradation.
- Mining sites are abandoned after excavation work is complete leaving deep scars and traces of over-burdening.

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- Deforestation due to mining have caused severe land degradation.
- Over irrigation is responsible for land degradation due to water logging leading to increase in salinity and alkalinity in the soil.
- Industrial effluents as waste have become a major source of land and water pollution in many parts of the country.

15. How can you conserve land?

- Practicing crop rotation and contour ploughing helps in maintaining soil fertility and nutrients from depletion.
- Planting trees on barren land stabilizes land degradation. Protecting green belts ensures that natural landscapes remain untouched by industrial expansion.
- Reducing plastic use and promoting composting keeps the land clean and productive.
- Designing sensitive areas like wetlands and grasslands as protected zones prevents them from being converted into commercial sites.

16. Do a comparative study between the two pie charts shown below (in page 5).

- Forest – Increased from 18.11% in 1960-1961 to over 23% (23.41%) by 2019-20.
- Net area sown – Remained relatively stable, showing a marginal increase from 45.26% (1960-61) to 45.64% (2019-20).
- Area under non-agricultural uses – Increased significantly due to urbanisation, infrastructure and industrial growth, rising from roughly 4.95% (1960-61) to nearly 9% (2019-20).
- Barren and unculturable wasteland – Decreased from 12% (1960-61) to 5.40% (2019-20), reflecting improvements in land reclamation.
- Culturable wasteland – Reduced from over 6% (1960-61) to roughly about 4% (2019-20) as this land was brought under cultivation.
- Permanent Pasture and Grazing land – Remained relatively low, showing minor fluctuates over the decades.

17. Why is soil an important resource?

Soil is the most important renewable natural resource. It is the medium of plant growth and supports different types of living organisms on the earth. The soil is a living system. It takes millions of years to form soil up to a few cm in depth. Soil is equally important. Soil also consists of organic (humus) and inorganic materials.

18. Name the factors for the formation of soil.

- Relief
- Parent rock or bedrock
- Climate
- Vegetation
- Other forms of life and time

Are important factors in the formation of soil.

19. Name the factors based on which soils are classified into different categories.

The factors based on which soils are classified into different categories are – colour, texture, structure, composition and fertility.

20. Write short notes on each of the categories of soil.

I. Alluvial Soils –

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- a) Formation – These have been deposited by three important Himalayan river systems– the Indus, the Ganga and the Brahmaputra.
- b) Region of availability –
 - The entire northern plains are made of alluvial soil.
 - These soils also extend in Rajasthan and Gujarat through a narrow corridor.
 - Alluvial soil is also found in the eastern coastal plains particularly in the deltas of the Mahanadi, the Godavari, the Krishna and the Kaveri rivers.
- c) Nutrients – Alluvial soils as a whole are very fertile. Mostly these soils contain adequate proportion of potash, phosphoric acid and lime.
- d) Crops grown – Sugarcane, Paddy, Wheat and other cereal and pulse crops.
- e) Other characteristics –
 - The alluvial soil consists of various proportions of sand, silt and clay.
 - According to their age alluvial soils can be classified as old alluvial (Bangar) and new alluvial (Khadar).
 - The Bangar soil has higher concentration of Kankar nodules than the Khadar. It has more fine particles and is more fertile than the Bangar.
 - Due to its high fertility, regions of alluvial soils are intensively cultivated and densely populated.

II. Black Soils –

- a) Formation –
 - It is believed that climatic condition along with the parent rock material are the important factors for the formation of black soil.
 - This type of soil is typical of the Deccan trap (*Basalt*) region spread over northwest Deccan plateau and is made up of lava flows.
- b) Region of availability – They cover the plateaus of Maharashtra, Saurashtra, Malwa, Madhya Pradesh and Chhattisgarh and extend in the south-east direction along the Godavari and the Krishna valleys.
- c) Nutrients –
 - They are rich in soil nutrients, such as calcium carbonate, magnesium, potash and lime.
 - These soils are generally poor in phosphoric contents.
- d) Crops grown – Cotton
- e) Other Characteristics –
 - These soils are black in colour and are also known as regur soils.
 - The black soils are made up of extremely fine i.e. clayey material. They are well-known for their capacity to hold moisture.
 - They develop deep cracks during hot weather, which helps in the proper aeration of the soil. These soils are sticky when wet and difficult to work on unless tilled immediately after the first shower or during the pre-monsoon period.

III. Red and Yellow Soils –

- a) Formation –
 - Red soil develops on crystalline igneous rocks.

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- These soils develop a reddish colour due to diffusion of iron in crystalline and metamorphic rocks. It looks yellow when it occurs in a hydrated form.
- b) Region of availability – Yellow and red soils are also found in the parts of the middle Ganga plain and along the piedmont zone of the Western Ghats.
- c) Crops grown – Cotton and millets

IV. Laterite Soils –

- a) Formation – These soils are formed due to lack of humus layer because of leaching and high temperature.
- b) Region of availability – Occur mostly in southern states, Western Ghats region of Maharashtra, Odisha, some parts of West Bengal and North-east regions.
- c) Nutrients –
 - Laterite Lateritic soils are mostly deep to very deep, acidic (pH<6.0), generally deficient in plant nutrients.
 - Where these soils support deciduous and evergreen forests, it is humus rich, but under sparse vegetation and in semi-arid environment, it is generally humus poor.
- d) Crops grown – Tea, Coffee, Cashew and Nuts.
- e) Other characteristics –
 - Laterite has been derived from the Latin word 'later' which means brick.
 - The laterite soil develops under tropical and subtropical climate with alternate wet and dry season.
 - They are prone to erosion and degradation due to their position on the landscape.
 - After adopting appropriate soil conservation techniques particularly in the hilly areas of Karnataka, Kerala and Tamil Nadu, this soil is very useful for growing tea and coffee.
 - Red laterite soils in Tamil Nadu, Andhra Pradesh and Kerala are more suitable for crops like cashew nut.

V. Arid Soils –

- a) Formation – Arid soils are formed due to lack of plant nutrients and moisture.
- b) Region of availability – Rajasthan
- c) Crops grown – Millets
- d) Other characteristics –
 - Arid soils range from red to brown in colour.
 - They are generally sandy in texture and saline in nature.
 - In some areas the salt content is very high and common salt is obtained by evaporating the water.
 - The lower horizons of the soil are occupied by *Kankar* because of the increasing calcium content downwards.
 - The *Kankar* layer formations in the bottom horizons restrict the infiltration of water.
 - After proper irrigation these soils become cultivable as has been in the case of western Rajasthan.

VI. Forest Soils –

- a) Formation – These soils are found in the hilly and mountainous areas where sufficient rain forests are available.
- b) Region of availability – Himalayan mountains

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- c) Nutrients – In the snow-covered areas of Himalayas, these soils experience denudation and are acidic with low humus content. The soils found in the lower parts of the valleys particularly on the river terraces and alluvial fans are fertile.
- d) Crops grown – Paddy, Maize, and Millets.
- e) Other characteristics – They are loamy and silty in valley sides and coarse grained in the upper slopes.

21. What is soil erosion?

The denudation of the soil cover and subsequent washing down is described as soil erosion.

22. Mention the reasons for soil erosion.

- [i] Deforestation - Removing trees eliminates root systems that bind soil together and takes away the leafy canopy that protects the ground from heavy rain.
- [ii] Over-grazing – Livestock consume too much vegetation, leaving the soil bare and compacting the earth with their hooves, which reduces water absorption and increases runoff.
- [iii] Mining – Large-scale excavation disturbs the land's natural structure and removes the topsoil layer, making the surrounding area highly susceptible to being washed or blown away.
- [iv] Construction – Clearing land for buildings and roads strips away protective plant cover and alters natural drainage patterns, leading to rapid sediment loss during storms.
- [v] Incorrect Method of Farming Practices – Techniques like "ploughing up and down a slope" create easy channels for water to carry soil away, rather than trapping it with contours or terraces.
- [vi] Landslides – The rapid downward movement of large masses of earth physically displaces vast amounts of topsoil and destabilizes the remaining landscape.

23. How can you conserve land?

- [i] Afforestation – Planting more and more trees
- [ii] Restricted grazing – Restricting livestock animals to a particular place to graze.
- [iii] Correct methods of farming should be practised –
 - Shelter belts
 - Contour ploughing
 - Terrace cultivation
 - Strip cropping

24. What do you mean by gullies and sheet erosion

Gullies – The running water cuts through the clayey soils and makes deep channels as gullies.

Sheet erosion – Sometimes water flows as a sheet over large areas down a slope. In such cases the top soil is washed away. This is known as sheet erosion.

25. What steps can be taken to control soil erosion in the hilly areas?

- [i] Growth of vegetation along the slopes.
- [ii] Proper farming like contour ploughing and terrace cultivation.